Wenjie Guan

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About me

I am a PhD student in Statistics at Cornell University, with research interests in machine learning theory, language modeling (transformer generalization), and high-dimensional statistics. I am experienced in theoretical analysis, probabilistic modeling, and large-scale simulation studies on clusters using Python/PyTorch and R.

Education

Cornell University – Ph.D. in Statistics Aug. 2024 – May. 2028 (expected)
GPA: 4.14. Advisor: Jelena Bradic. *Ithaca, NY*

University of Washington – M.S. in Statistics Sept. 2022 – Jun. 2024
GPA: 3.95/4.0 *Seattle, WA*

Fudan University – B.S. in Mathematics and Applied Math Sept. 2018 – Jun. 2022
GPA: 3.70/4.0 (with major GPA 3.88) *Shanghai, China*

Work Experience

Handshake MoVE program Jul. 2025 – Feb. 2026
Math Fellow and Reviewer – part time *Remote*

- My duty is to write graduate-level math and statistics prompts that LLM models fail to answer or prove correctly, evaluate their reasoning capabilities and explain where they fail, and provide the step-by-step solutions to the prompts. Also I review the prompts provided by fellows and move the good ones to production.

Publications and Working Papers

W. Guan, J. Bradic. When Symbol Names Should Not Matter: A Logistic Theory of Fresh-Symbol Classification, submitted to *the Conference on Neural Information Processing Systems (NeurIPS)*, 2026. ArXiv.

W. Guan, J. Ma. Network Discovery and Clustering with Block-structured Gaussian Graphical Model, submitted to *Journal of Computational and Graphical Statistics*, 2026.

W. Guan, Y. Zhang, D. Zhou, T. Cai, A. Giessing. Efficient Inference on High-Dimensional Logistic Regression under Class Imbalance, under preparation.

Research Experience

Generalization of Transformers on Classification Tasks May 2025 – Now
Advisor: Jelena Bradic, professor at Cornell University *Ithaca, NY*

- Developed a theoretical framework for unseen-symbol generalization in one-layer transformers by reducing ridge-regularized transformer training to kernel logistic regression in the infinite-width regime.
- Introduced a collision-graph perturbation framework characterizing how finite-sample token collisions affect generalization, and derived probabilistic margin-transfer guarantees for template-based classification tasks.
- Conducted simulation studies validating the role of attention regularization and dataset collision geometry in fresh-symbol generalization.

Inference for High-dimensional Binary Classification with Class Imbalance Jun. 2023 – May 2025
Advisor: Alexander Giessing, now assistant professor at National University of Singapore *Seattle, WA*

- Proposed the bias correction procedure with a convex program based on bias-variance trade-off.
- Proved the consistency and asymptotic efficiency of the estimator in high-dimensional settings and rare events.
- Implmented the algorithm and conducted simulation studies. Our method outperforms the existing inference methods by having smaller variance and better normality under the logistic model with imbalanced sample.
- Currently working on the paper and package, and applying our method on type II diabetes (T2D) data.

Clustered Graphical Modeling for Microbiome Data Oct. 2022 – Mar. 2026
Advisor: Jing Ma, from Fred Hutchinson Cancer Center *Seattle, WA*

- Built a hierarchical Gaussian graphical model with adjacency matrix following a stochastic block model.
- Estimated the precision matrix, model parameters and cluster membership using the variational EM method.
- Implemented the algorithm on the simulated data. It outperformed the existing clustered graphical Lasso algorithms in network discovery and clustering when the network is generated from a stochastic block model.

- Applied our method on S&P 100 and microbiome data, our model achieved higher test accuracy and better interpretation than existing methods.

Awards

National First Prize | 12^{th} *National College Mathematics Competition (Final)* May. 2021

Finalist | 12^{th} *S.-T. Yau College Student Mathematics Contest – Probability and Statistics* May. 2021

Skills

Languages R (proficient), Python (Numpy, Pandas, Sklearn, Pytorch), LaTeX, C++.

ML/AI PyTorch, scikit-learn, transformers, statistical learning.

Scientific Computing NumPy, Pandas, SciPy.

Research high-dimensional statistics, asymptotic theory, optimization, probabilistic modeling.

Tools Git, Linux, LaTeX, SLURM.